Power-Optimal Inverter Chain Design Under Elmore Delay Constraints

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1 Introduction

1.1 Motivation

Modern VLSI digital circuits demand increasingly stringent performance and energy efficiency requirements, particularly as transistor scaling continues to push designs toward higher density and faster clock speeds. A fundamental component in high-speed digital design is the use of inverter chains (buffer chains) to drive large capacitive loads, where improper sizing or staging can lead to excessive delay or unnecessary power consumption. While classical design heuristics such as fixed stage effort or logical effort methods provide useful guidelines, they do not always yield globally optimal solutions under complex constraints involving wire parasitics, transistor sizing limits, and timing requirements. The goal of this work is to develop a principled optimization framework for inverter chain design that minimizes dynamic power consumption subject to an Elmore delay constraint. We approach this problem through the lens of convex optimization by reformulating the inherently non-convex delay expressions into a geometric programming (GP) framework via logarithmic transformation. This enables efficient computation of globally optimal inverter sizes and interconnect lengths while accounting for physical constraints. Through this formulation, we aim to bridge classical circuit design intuition with modern optimization techniques and provide a scalable method for a energy-efficient buffer chain design.

1.2 Previous Works

1.3 Intended Contributions

1. Optimization-Based Formulation: We formulate the inverter chain design problem as a power minimization problem with Elmore delay, sizing, and interconnect constraints.

2. Geometric Programming Reformulation: We show that the non-convex delay constraint can be expressed as a posynomial and transformed into a convex constraint via logarithmic variable substitution.

3. Convexity Verification: We provide a formal justification that the transformed delay constraint is convex by leveraging the log-sum-exp structure and verifying all terms satisfy geometric programming requirements.

4. Solver Implementation: We implement the resulting formulation in a convex optimization framework (e.g., CVXPY) to compute optimal inverter sizing and interconnect parameters.

5. Design Insight: We analyze the trade-offs between delay, sizing, and power, and compare the optimized solutions to classical inverter chain design heuristics.

1.4 Organization of the Paper

2 Statement of the Problem

2.1 Primal Formulation

2.1.1 Physical Interpretation of the Inverter Chain

We consider a chain of N CMOS inverters used to drive a fixed load capacitance C_L. Such inverter chains, commonly referred to as buffer chains, are fundamental building blocks in high-speed digital circuits, where they are used to drive large capacitive loads while meeting strict timing constraints.

Each stage in the inverter chain is characterized by two primary design variables:

- $W_i > 0$: the effective transistor width (or sizing factor) of inverter i
- $l_i > 0$: the interconnect length between inverter i and inverter $i+1$

The objective is to determine the optimal values of $\{W_i\}_{i=1}^N$ and $\{l_i\}_{i=1}^N$ that minimize dynamic power consumption while satisfying a delay constraint. Each inverter contributes capacitance from two sources:

Gate (Input) Capacitance The input capacitance of inverter i is proportional to its size:

$$C_{in,i} = W_i C_{g0}$$

where C_{g0} represents the gate capacitance of a unit-sized inverter.

Parasitic Capacitance Each inverter also has intrinsic parasitic capacitance due to diffusion regions:

$$C_{p,i} = W_i C_{p0}$$

where C_{p0} is the parasitic capacitance of a unit inverter.

These capacitances contribute to both dynamic power consumption and delay. The wires connecting inverter stages introduce additional parasitics:

Interconnect Modeling

- Resistance:

$$R_{wire,i} = r\, l_i$$

- Capacitance:

$$C_{wire,i} = c\, l_i$$

where r and c denote per-unit-length resistance and capacitance, respectively. These parameters capture the physical effects of interconnect scaling and play a critical role in delay modeling.

Inverter Resistance The effective output resistance of inverter i is inversely proportional to its size:

$$R_i = \frac{R_0}{W_i}$$

where R_0 is the resistance of a unit inverter. Increasing the transistor width reduces resistance, improving delay at the expense of increased capacitance.

Dynamic Power Objective The dynamic power consumption of a digital circuit is given by:

$$P_{dyn} = \alpha f V_{DD}^2 C_{tot}$$

where α is the switching activity factor, f is the clock frequency, and V_{DD} is the supply voltage. These quantities are treated as constants in our formulation.

Thus, minimizing dynamic power reduces to minimizing the total switched capacitance C_{tot}.

The total capacitance in the inverter chain consists of:

- Parasitic capacitance of each inverter
- Gate capacitance of downstream inverters
- Interconnect capacitance
- Final load capacitance C_L

This yields:

$$C_{tot} = \sum_{i=1}^{N} W_i C_{p0} + \sum_{i=1}^{N-1} W_{i+1} C_{g0} + \sum_{i=1}^{N} c\, l_i + C_L$$

Since C_L is constant, it does not affect the optimization and may be omitted from the objective.

Elmore Delay Constraint To ensure correct circuit operation, the total propagation delay must not exceed a specified clock period T_{clk}. We model delay using the Elmore delay approximation, which expresses delay as a weighted sum of downstream capacitances. The delay contribution of each stage includes interactions between inverter resistance and downstream capacitances, as well as interconnect effects.

For stages $i = 1, \ldots, N-1$, the delay contribution is:

$$d_i = R_0 C_{g0} \frac{W_{i+1}}{W_i} + R_0 C_{p0} + R_0 c \frac{l_i}{W_i} + r C_{g0} l_i W_{i+1} + r C_{p0} l_i W_i + r c l_i^2$$

The final stage differs since it drives the external load C_L:

$$d_N = R_0 C_L \frac{1}{W_N} + R_0 C_{p0} + R_0 c \frac{l_N}{W_N} + r C_L l_N + r C_{p0} l_N W_N + r c l_N^2$$

The total delay constraint is therefore:

$$\sum_{i=1}^{N-1} d_i + d_N \leq T_{clk}$$

Design Constraints To ensure physically realizable designs, we impose:

$$W_{min} \leq W_i \leq W_{max}, \quad W_i > 0, \quad l_i > 0$$

2.1.2 Primal Statement

Geometric Programming Reformulation The original formulation is non-convex due to multiplicative and ratio terms such as:

$$\frac{W_{i+1}}{W_i}, \quad W_i l_i, \quad l_i^2$$

To enable efficient optimization, we reformulate the problem using geometric programming techniques.

Logarithmic Variable Transformation We introduce the change of variables:

$$x_i = \log W_i, \qquad y_i = \log l_i$$

which implies:

$$W_i = e^{x_i}, \qquad l_i = e^{y_i}$$

This transformation is bijective over the positive orthant, ensuring no loss of generality.

Transformation of Nonlinear Terms Under this transformation, multiplicative and ratio terms become exponentials of affine expressions:

$$\frac{W_{i+1}}{W_i} = e^{x_{i+1}-x_i}, \quad W_i l_i = e^{x_i+y_i}, \quad l_i^2 = e^{2y_i}$$

Thus, each term in the delay expressions can be written in the form:

$$c_k e^{a_k^T z}, \quad \text{where } z = [x_1, \ldots, x_N, y_1, \ldots, y_N]^T$$

Convexity of the Log-Sum-Exp Function Define:

$$f(x) = \log\left(\sum_{k=1}^{n} e^{x_k}\right)$$

Let $z_k = e^{x_k}$ and $z = [z_1, \ldots, z_n]^T$. Then:

$$\nabla^2 f(x) = \frac{1}{\mathbf{1}^T z}\mathrm{diag}(z) - \frac{1}{(\mathbf{1}^T z)^2} z z^T$$

For any $v \in \mathbb{R}^n$:

$$v^T \nabla^2 f(x) v = \frac{1}{(\mathbf{1}^T z)^2}\left[(\mathbf{1}^T z)\left(\sum_{i=1}^{n} z_i v_i^2\right) - \left(\sum_{i=1}^{n} z_i v_i\right)^2\right]$$

By the Cauchy–Schwarz inequality:

$$\left(\sum_{i=1}^{n} z_i v_i\right)^2 \leq (\mathbf{1}^T z)\left(\sum_{i=1}^{n} z_i v_i^2\right)$$

Thus:

$$v^T \nabla^2 f(x) v \geq 0 \quad \forall v$$

which proves that $f(x)$ is convex.

Convexity of the Delay Constraint The total delay can be written as:

$$D(x, y) = \sum_k c_k e^{a_k^T z}$$

Taking logarithms:

$$\log D(x, y) = \log\left(\sum_k e^{a_k^T z + \log c_k}\right)$$

which is a log-sum-exp function and therefore convex. Hence, the constraint:

$$D(x, y) \leq T_{clk}$$

is equivalent to the convex constraint:

$$\log D(x, y) \leq \log T_{clk}$$

Final Convex Optimization Problem The optimization problem becomes:

$$\min_{x,y} \quad \log\left(C_{p0}e^{x_1} + \sum_{i=2}^{N}(C_{g0} + C_{p0})e^{x_i} + c\sum_{i=1}^{N} e^{y_i}\right) \tag{1}$$

$$\text{s.t.} \quad \log\left(\sum_{i=1}^{N-1} d_i(x,y) + d_N(x,y)\right) \leq \log T_{clk} \tag{2}$$

$$\log W_{min} \leq x_i \leq \log W_{max}, \quad \forall i$$

$$\log \epsilon \leq y_i \leq \log l_{max}, \quad \forall i$$

Strong Duality and Slater's Condition The problem is convex since:

- The objective is a log-sum-exp function, hence convex.
- The delay constraint is convex.
- The bound constraints are affine.

Let the feasible set be:

$$\mathcal{D} = \{(x,y) \mid \log W_{min} \leq x_i \leq \log W_{max},\ \log \epsilon \leq y_i \leq \log l_{max}\}$$

The relative interior of $\mathcal{D}$ is:

$$\text{relint}(\mathcal{D}) = \{(x,y) \mid \log W_{min} < x_i < \log W_{max},\ \log \epsilon < y_i < \log l_{max}\}$$

To verify Slater's condition, we must show that there exists $(x^*, y^*) \in \text{relint}(\mathcal{D})$ such that:

$$\log\left(\sum_{i=1}^{N-1} d_i(x^*,y^*) + d_N(x^*,y^*)\right) < \log T_{clk}$$

Such a point exists by choosing:

- y_i^* sufficiently small (close to $\log \epsilon$), which reduces all terms involving e^{y_i},
- x_i^* strictly within bounds and arranged to balance exponential terms.

Since all delay terms are positive and continuous, and can be made sufficiently small relative to T_{clk}, a strictly feasible point exists.

Therefore, Slater's condition holds:

$$\exists (x^*, y^*) \in \text{relint}(\mathcal{D})$$

such that

$$f_i(x^*, y^*) < 0$$

Conclusion Since the problem is convex and Slater's condition holds, strong duality follows:

$$p^* - d^* = 0$$

Thus, the duality gap is zero, and the Karush-Kuhn-Tucker (KKT) conditions are necessary and sufficient for optimality.

2.2 Dual Formulation

Define

$$S_0(x, y) = C_{p0}e^{x_1} + \sum_{i=1}^{N}(C_{g0} + C_{p0})e^{x_i} + c\sum_{i=1}^{N} e^{y_i},$$

and

$$S_1(x, y) = \sum_{i=1}^{N-1} d_i(x, y) + d_N(x, y),$$

where, for $i = 1, \ldots, N-1$
The Lagrangian is

$$\begin{aligned}\mathcal{L}(x, y, \lambda, \alpha, \beta, \gamma, \delta) = {} & \log S_0(x, y) + \lambda\big(\log S_1(x, y) - \log T_{clk}\big) \\ & + \sum_{i=1}^{N} \alpha_i(\log W_{\min} - x_i) + \sum_{i=1}^{N} \beta_i(x_i - \log W_{\max}) \\ & + \sum_{i=1}^{N} \gamma_i(\log \varepsilon - y_i) + \sum_{i=1}^{N} \delta_i(y_i - \log l_{\max}).\end{aligned} \tag{3}$$

Using the conjugate of log-sum-exp

$$\log\left(\sum_j e^{s_j}\right) = \max_{\omega_j \geq 0,\ \sum_j \omega_j = 1} \left(\sum_j \omega_j s_j - \sum_j \omega_j \log \omega_j\right),$$

introduce nonnegative weights $\theta_0, \theta_i, \phi_i$ such that

$$\theta_0 + \sum_{i=1}^{N} \theta_i + \sum_{i=1}^{N} \phi_i = 1.$$

Then

$$\begin{aligned}\log S_0(x, y) = \max_{\theta, \phi} \Bigg\{ & \theta_0(\log C_{p0} + x_1) + \sum_{i=1}^{N} \theta_i\big(\log(C_{g0} + C_{p0}) + x_i\big) \\ & + \sum_{i=1}^{N} \phi_i(\log c + y_i) - \theta_0 \log \theta_0 - \sum_{i=1}^{N} \theta_i \log \theta_i - \sum_{i=1}^{N} \phi_i \log \phi_i \Bigg\}.\end{aligned} \tag{4}$$

Using the scaled conjugate log-sum-exp

$$\lambda \log\left(\sum_j e^{s_j}\right) = \max_{\mu_j \geq 0,\ \sum_j \mu_j = \lambda} \left(\sum_j \mu_j s_j - \sum_j \mu_j \log \mu_j + \lambda \log \lambda\right),$$

introduce nonnegative variables $\eta_{i,k}$, $k = 1, \ldots, 6$, such that

$$\sum_{i=1}^{N} \sum_{k=1}^{6} \eta_{i,k} = \lambda.$$

For $i = 1, \ldots, N-1$, let

$$\eta_{i,1} \leftrightarrow R_0 C_{g0} e^{x_{i+1} - x_i}, \quad \eta_{i,2} \leftrightarrow R_0 C_{p0}, \quad \eta_{i,3} \leftrightarrow R_0 c e^{y_i - x_i},$$

$$\eta_{i,4} \leftrightarrow r C_{g0} e^{y_i + x_{i+1}}, \quad \eta_{i,5} \leftrightarrow r C_{p0} e^{y_i + x_i}, \quad \eta_{i,6} \leftrightarrow r c e^{2y_i},$$

and for $i = N$,

$$\eta_{N,1} \leftrightarrow R_0 C_L e^{-x_N}, \quad \eta_{N,2} \leftrightarrow R_0 C_{p0}, \quad \eta_{N,3} \leftrightarrow R_0 c e^{y_N - x_N},$$

$$\eta_{N,4} \leftrightarrow r C_L e^{y_N}, \quad \eta_{N,5} \leftrightarrow r C_{p0} e^{y_N + x_N}, \quad \eta_{N,6} \leftrightarrow r c e^{2y_N}.$$

Then

$$\begin{aligned} \lambda \log S_1(x,y) = \max_{\eta \geq 0,\ \sum \eta = \lambda} \Bigg\{ & \sum_{i=1}^{N-1} \sum_{k=1}^{6} \eta_{i,k} \big(\log a_{i,k} + m_{i,k}(x,y)\big) \\ & + \sum_{k=1}^{6} \eta_{N,k} \big(\log a_{N,k} + m_{N,k}(x,y)\big) - \sum_{i=1}^{N} \sum_{k=1}^{6} \eta_{i,k} \log \eta_{i,k} + \lambda \log \lambda \Bigg\}. \end{aligned} \tag{5}$$

Substituting (4) and (5) into (3), the resulting expression is affine in x and y. Finite infimum over x and y requires the coefficients of each x_i and y_i to vanish. From x_1,

$$\theta_0 + \theta_1 - \eta_{1,1} - \eta_{1,3} + \eta_{1,5} - \alpha_1 + \beta_1 = 0.$$

From x_i, $i = 2, \ldots, N-1$,

$$\theta_i + \eta_{i-1,1} + \eta_{i-1,4} - \eta_{i,1} - \eta_{i,3} + \eta_{i,5} - \alpha_i + \beta_i = 0.$$

From x_N,

$$\theta_N + \eta_{N-1,1} + \eta_{N-1,4} - \eta_{N,1} - \eta_{N,3} + \eta_{N,5} - \alpha_N + \beta_N = 0.$$

From y_i, $i = 1, \ldots, N$,

$$\phi_i + \eta_{i,3} + \eta_{i,4} + \eta_{i,5} + 2\eta_{i,6} - \gamma_i + \delta_i = 0.$$

Then the dual function is

$$
\begin{aligned}
g = {} & \theta_0 \log \frac{C_{p0}}{\theta_0} + \sum_{i=1}^{N} \theta_i \log \frac{C_{g0} + C_{p0}}{\theta_i} + \sum_{i=1}^{N} \phi_i \log \frac{c}{\phi_i} \\
& + \sum_{i=1}^{N-1} \left[\eta_{i,1} \log \frac{R_0 C_{g0}}{\eta_{i,1}} + \eta_{i,2} \log \frac{R_0 C_{p0}}{\eta_{i,2}} + \eta_{i,3} \log \frac{R_0 c}{\eta_{i,3}} + \eta_{i,4} \log \frac{r C_{g0}}{\eta_{i,4}} + \eta_{i,5} \log \frac{r C_{p0}}{\eta_{i,5}} + \eta_{i,6} \log \frac{rc}{\eta_{i,6}} \right] \\
& + \eta_{N,1} \log \frac{R_0 C_L}{\eta_{N,1}} + \eta_{N,2} \log \frac{R_0 C_{p0}}{\eta_{N,2}} + \eta_{N,3} \log \frac{R_0 c}{\eta_{N,3}} + \eta_{N,4} \log \frac{r C_L}{\eta_{N,4}} + \eta_{N,5} \log \frac{r C_{p0}}{\eta_{N,5}} + \eta_{N,6} \log \frac{rc}{\eta_{N,6}} \\
& + \lambda \log \lambda - \lambda \log T_{clk} + \sum_{i=1}^{N} \alpha_i \log W_{\min} - \sum_{i=1}^{N} \beta_i \log W_{\max} + \sum_{i=1}^{N} \gamma_i \log \varepsilon - \sum_{i=1}^{N} \delta_i \log l_{\max}.
\end{aligned}
$$

Therefore, the dual problem is

$$
\begin{aligned}
\max_{\theta,\phi,\eta,\lambda,\alpha,\beta,\gamma,\delta} \quad & g \\
\text{s.t.} \quad & \theta_0 \geq 0, \qquad \theta_i \geq 0, \qquad \phi_i \geq 0, \qquad i = 1, \ldots, N, \\
& \eta_{i,k} \geq 0, \qquad i = 1, \ldots, N, \quad k = 1, \ldots, 6, \\
& \alpha_i, \beta_i, \gamma_i, \delta_i \geq 0, \qquad i = 1, \ldots, N, \\
& \theta_0 + \sum_{i=1}^{N} \theta_i + \sum_{i=1}^{N} \phi_i = 1, \\
& \sum_{i=1}^{N} \sum_{k=1}^{6} \eta_{i,k} = \lambda, \\
& \theta_0 + \theta_1 - \eta_{1,1} - \eta_{1,3} + \eta_{1,5} - \alpha_1 + \beta_1 = 0, \\
& \theta_i + \eta_{i-1,1} + \eta_{i-1,4} - \eta_{i,1} - \eta_{i,3} + \eta_{i,5} - \alpha_i + \beta_i = 0, \quad i = 2, \ldots, N-1, \\
& \theta_N + \eta_{N-1,1} + \eta_{N-1,4} - \eta_{N,1} - \eta_{N,3} + \eta_{N,5} - \alpha_N + \beta_N = 0, \\
& \phi_i + \eta_{i,3} + \eta_{i,4} + \eta_{i,5} + 2\eta_{i,6} - \gamma_i + \delta_i = 0, \quad i = 1, \ldots, N.
\end{aligned}
$$

2.3 KKT Conditions

- Primal feasibility:

$$
\begin{aligned}
& \log S_1(x, y) \leq \log T_{clk} \\
& \log W_{\min} \leq x_i \leq \log W_{\max}, \forall i \\
& \log \epsilon \leq y_i \leq \log l_{\max}, \forall i
\end{aligned}
$$

- Dual feasibility:

$$
\lambda \geq 0, \quad \alpha_i \geq 0, \quad \beta_i \geq 0, \quad \gamma_i \geq 0, \quad \delta_i \geq 0, \quad \forall i
$$

- Complementary slackness:

$$
\begin{aligned}
&\lambda(\log S_1(x,y) - \log T_{clk}) = 0 \\
&\alpha_i(\log W_{\min} - x_i) = 0, \forall i \\
&\beta_i(x_i - \log W_{\max}) = 0, \forall i \\
&\gamma_i(\log \epsilon - y_i) = 0, \forall i \\
&\delta_i(y_i - \log l_{\max}) = 0, \forall i
\end{aligned}
$$

- Stationarity:

$$
\begin{aligned}
\frac{\partial \mathcal{L}}{\partial x_i} &= \frac{\partial}{\partial x_i} \log S_0(x,y) + \lambda \frac{\partial}{\partial x_i}(\log S_1(x,y) - \log T_{clk}) - \alpha_i + \beta_i \\
&= \frac{C_{p0}e^{x_1} + (C_{g0} + C_{p0})e^{x_i}}{S_0(x,y)} \\
&+ \frac{\lambda(R_0 C_{g0} e^{x_i - x_{i-1}} + rC_{go}e^{y_{i-1}+x_i} - R_0 C_{g0} e^{x_{i+1}-x_i} - R_0 c E^{y_i - x_i} + rC_{p0}e^{y_i + x_i})}{S_1(x,y)} - \alpha_i + \beta_i = 0
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial \mathcal{L}}{\partial y_i} &= \frac{\partial}{\partial y_i} \log S_0(x,y) + \lambda \frac{\partial}{\partial y_i}(\log S_1(x,y) - \log T_{clk}) - \gamma_i + \delta_i \\
&= \frac{ce^{y_i}}{S_0(x,y)} + \frac{\lambda(R_0 c e^{y_i - x_i} + rC_{g0}e^{y_i + x_{i+1}} + rC_{p0}e^{y_i + x_i} + 2rce^{2y_i})}{S_1(x,y)} - \gamma_i + \delta_i = 0
\end{aligned}
$$

3 Approaches

4 Results And Conclusion

4.1 Results

4.2 Conclusion

5 References